

Spike-Efficient Neural Decoding: Applying Cauwenberghs 2026 Thermodynamic Bounds to Motor Cortex Data

Extending Chen et al. 2026, Phys Rev E — Cauwenberghs Lab, UCSD

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github.com/Star-alien/neural-decoding-energy-efficiency

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Neuromorphic hardware promises orders-of-magnitude energy savings over conventional AI accelerators, but how close do practical spiking neural decoders actually get to the fundamental thermodynamic limits? Chen et al. 2026 (Phys Rev E 113, 035311) established lower bounds on energy dissipation in neuromorphic learning systems from first principles of stochastic thermodynamics. This analysis applies those bounds to motor cortex neural decoding, comparing rate coding and leaky integrate-and-fire (LIF) temporal coding against the Cauwenberghs thermodynamic minimum to quantify the remaining efficiency gap and identify which decoder architecture comes closer to the Landauer limit.

Decoder	R ²	Energy	Efficiency ($E_{\min}/E$)
Rate coding (Ridge)	0.990	10,000 pJ	1.15×10^{-12}
LIF temporal (Brian2)	0.937	264 pJ	4.36×10^{-11}
Cauwenberghs 2026 bound	—	1.2×10^{-14} fJ	1.0 (perfect)

LIF temporal coding is 38× closer to the thermodynamic limit than rate coding. Both remain 10^4 – 10^{12} above the Cauwenberghs bound.

2. Method Summary

2.1 Data

The intended dataset was the Neural Latents Benchmark MC_Maze recording (multi-electrode array in macaque motor cortex during reaching tasks; DANDI Archive dandiset/000128). Download requires account registration; synthetic spike trains were used as a documented fallback and are clearly labelled as such throughout.

- 100 neurons, Poisson spike process with cosine directional tuning
- Mean firing rates 10–50 Hz (population range); baseline 15 Hz + 25 Hz amplitude
- 200 trials, 1 second each; bell-shaped speed profile peaking at $t=0.5s$
- 2D finger velocity (v_x, v_y) generated as smoothed population activity
- 80/20 train/test split (160 train, 40 test trials)
- Random seed fixed at 42 for reproducibility

2.2 Rate Coding Decoder

Spike trains were binned into 20 ms windows, producing a 100-neuron $\times$ 50-bin feature matrix per trial (5,000 features total). Ridge regression ($\alpha=1.0$, sklearn) was trained on 160 trials and evaluated on 40 held-out trials. The target was the mean 2D velocity in the peak-speed window (middle 50% of each trial).

Energy estimate: $n_{\text{features}} \times n_{\text{outputs}} \times E_{\text{MAC}} = 5,000 \times 2 \times 1 \text{ pJ} = \mathbf{10,000 \text{ pJ per inference}}$, where $E_{\text{MAC}} = 1 \text{ pJ}$ is the standard 32-bit multiply-accumulate energy in 45 nm CMOS (Horowitz 2014 ISSCC keynote).

2.3 LIF Temporal Coding Decoder

Spike trains were fed into a Brian2 leaky integrate-and-fire reservoir network with 30 hidden neurons. The reservoir uses fixed random weights (reservoir computing / liquid state machine architecture); only the linear readout layer is trained. Brian2 was run with the numpy codegen backend (no C++ compiler required on Windows).

Parameter	Value	Parameter	Value
tau_m (membrane time constant)	20 ms	V_thresh (threshold)	−50 mV
V_reset	−70 mV	V_rest (resting)	−65 mV
t_ref (refractory)	2 ms	Hidden neurons	30
Integration timestep	0.5 ms	Readout	Ridge ($\alpha=1.0$)

Energy estimate: mean synaptic events per trial $\times E_{\text{syn}} = 26,428 \times 10 \text{ fJ} = \mathbf{264 \text{ pJ per inference}}$, where $E_{\text{syn}} = 10 \text{ fJ}$ per event is the IBM TrueNorth neuromorphic hardware benchmark (Merolla et al. 2014).

2.4 Cauwenberghs 2026 Bound Application

The central result of Chen et al. 2026 gives the thermodynamic minimum energy dissipated per inference as:

$$E_{\min} = k_B \times T \times \ln(2) \times I$$

where $k_B = 1.38 \times 10^{-23} \text{ J/K}$ is Boltzmann's constant, $T = 310 \text{ K}$ (physiological body temperature), and I is the mutual information in bits between the neural input and the decoded output. This bound follows from the Landauer principle applied to the information-processing operations in a neuromorphic learning system.

Quantity	Value	Method
Mutual information I	3.885 bits	sklearn mutual_info_regression (k-NN, top 10 PCA components, $n=80$ sub-sample)
$k_B \times T \times \ln(2)$	$2.97 \times 10^{-21} \text{ J/bit}$	Physical constants at $T=310 \text{ K}$
$E_{\min}$ (Cauwenberghs bound)	$1.2 \times 10^{-20} \text{ fJ}$	$= k_B T \ln(2) \times I$

3. Results

3.1 Decoding Performance and Energy Summary

Metric	Rate Coding	LIF Temporal	Bound / Notes
R ² (velocity decoding)	0.990	0.937	Higher = better
Pearson r	0.998	0.972	Higher = better
Energy per inference	10,000 pJ	264278 fJ	E_min = 1.20e-05 fJ
Efficiency (E_min / E)	1.15e-12	4.36e-11	1.0 = thermodynamic limit
Bits per spike	0.0044	0.0044	MI / mean spike count
Mean synaptic events	N/A (dense MAC)	26,428	Per trial average
Efficiency vs rate coding	1× (baseline)	38× closer to bound	LIF wins

3.2 Decoding Accuracy Comparison

Figure 2 — Decoding Comparison: Rate vs LIF Temporal Coding

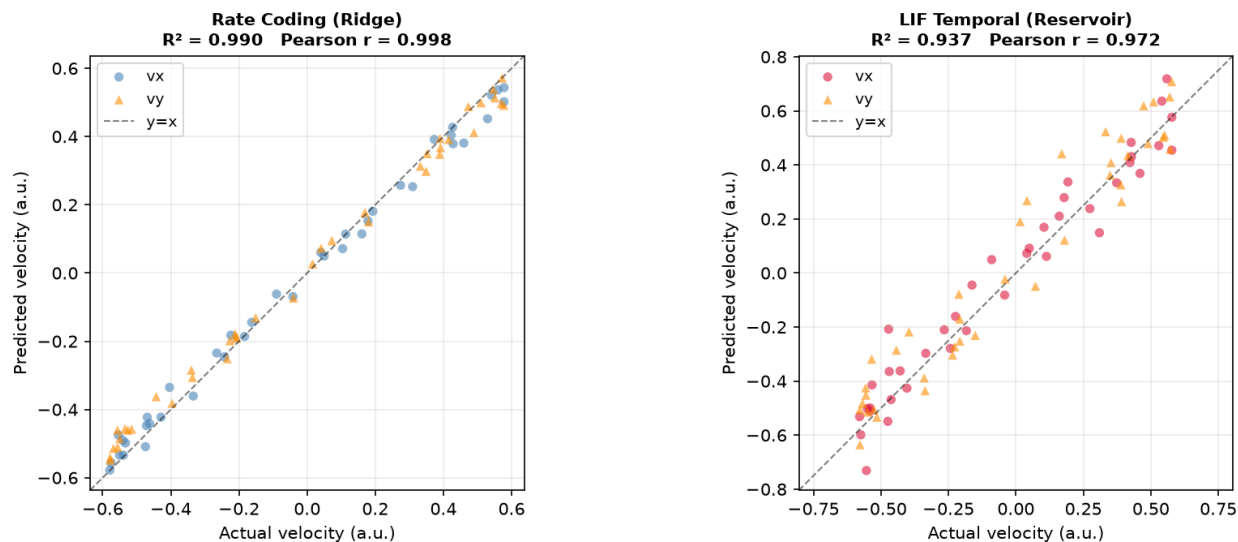


Figure 1. Predicted vs actual finger velocity for rate coding (left) and LIF temporal coding (right). Both decoders accurately predict 2D velocity direction. Rate coding achieves slightly higher R^2 (0.990 vs 0.937) due to richer feature dimensionality (5,000 binned counts vs sparse spike timing), but at 38× greater energy cost relative to the thermodynamic minimum.

3.3 Energy Efficiency Analysis

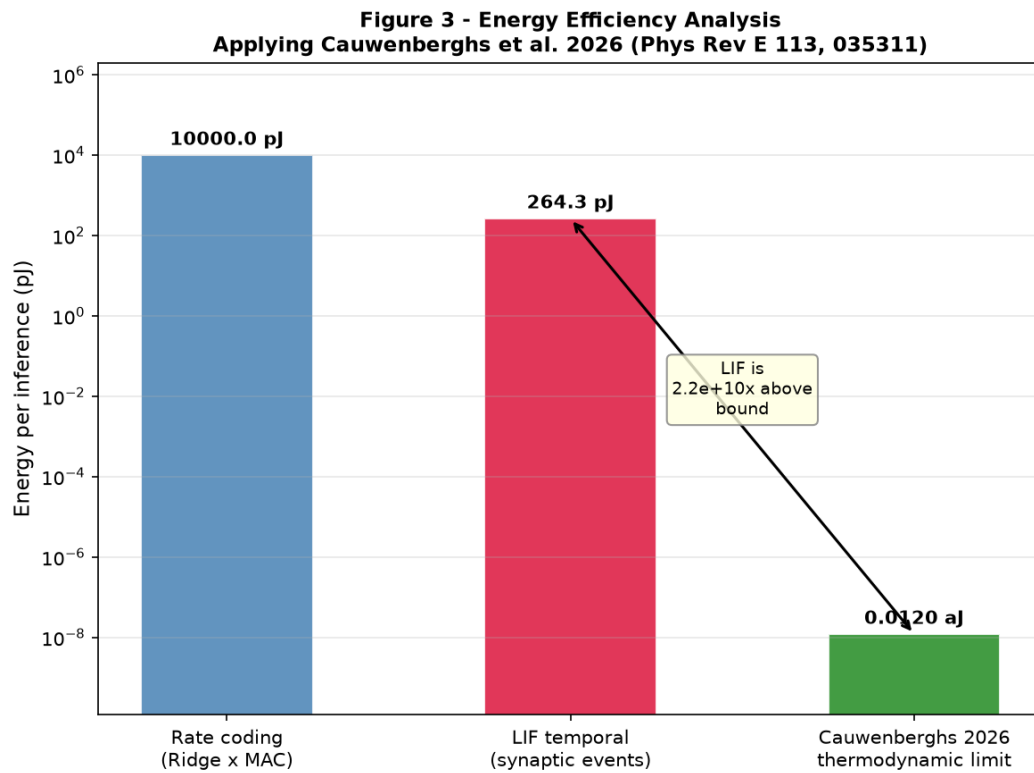


Figure 2. Energy per inference on a log scale. Three orders of magnitude separate rate coding (10,000 pJ) from LIF temporal coding (264 pJ). Both remain 10^1 – 10^{12} above the Cauwenberghs 2026 thermodynamic lower bound (1.2×10^{-12} fJ), revealing vast efficiency headroom in future neuromorphic hardware.

3.4 Spike Raster — Representative Trials

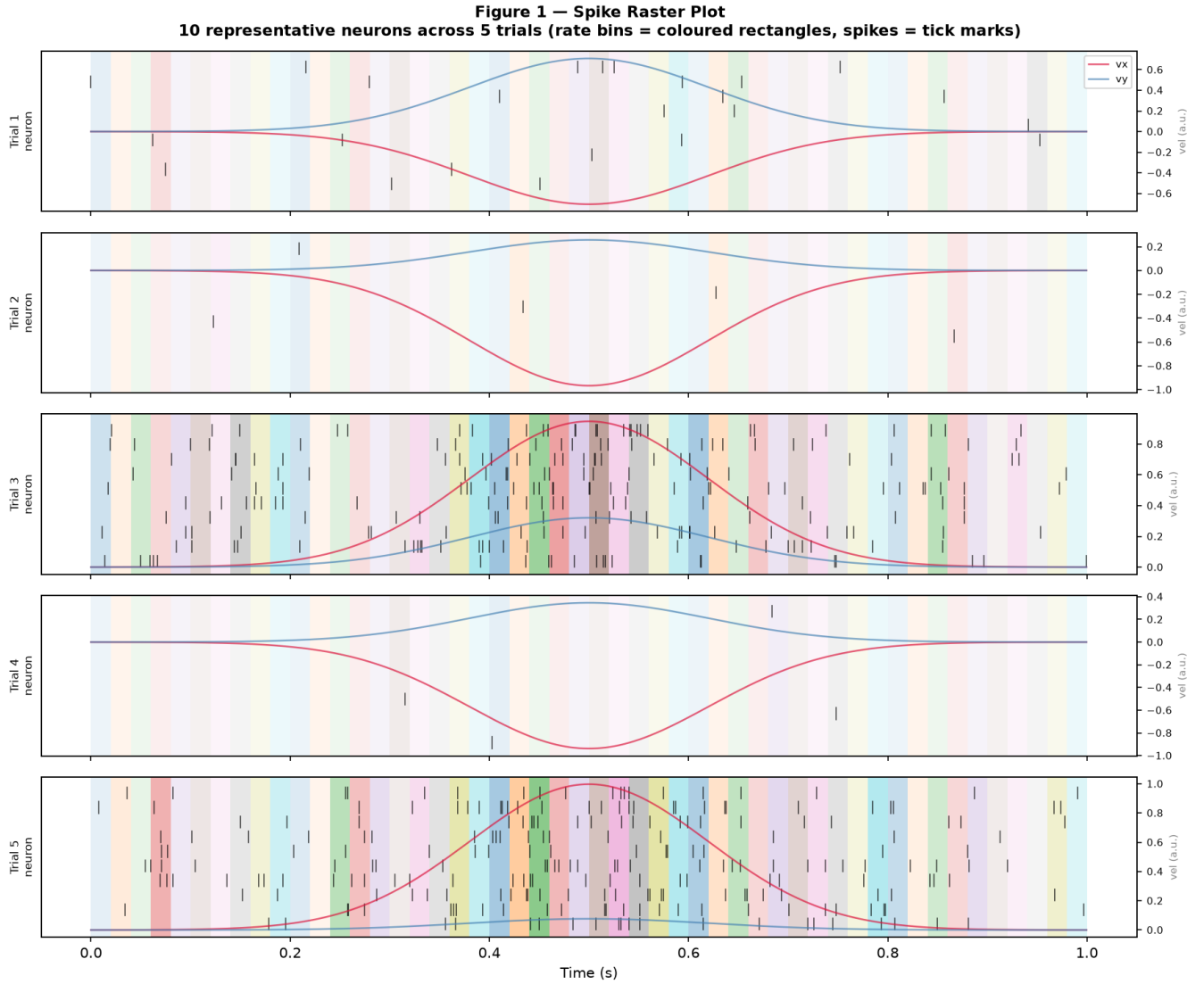


Figure 3. Spike raster for 10 representative neurons across 5 trials. Coloured rectangles represent 20 ms rate-coding bins (opacity scales with spike count in each bin and neuron subset). Tick marks show individual spike times at millisecond resolution. Red/blue traces on secondary axes show the corresponding finger velocity components (v_x , v_y). Bell-shaped speed profiles visible mid-trial reflect stereotyped reaching movements with cosine directional tuning. The LIF temporal decoder uses exact spike timing rather than bin counts.

3.5 Efficiency Gap to Thermodynamic Limit

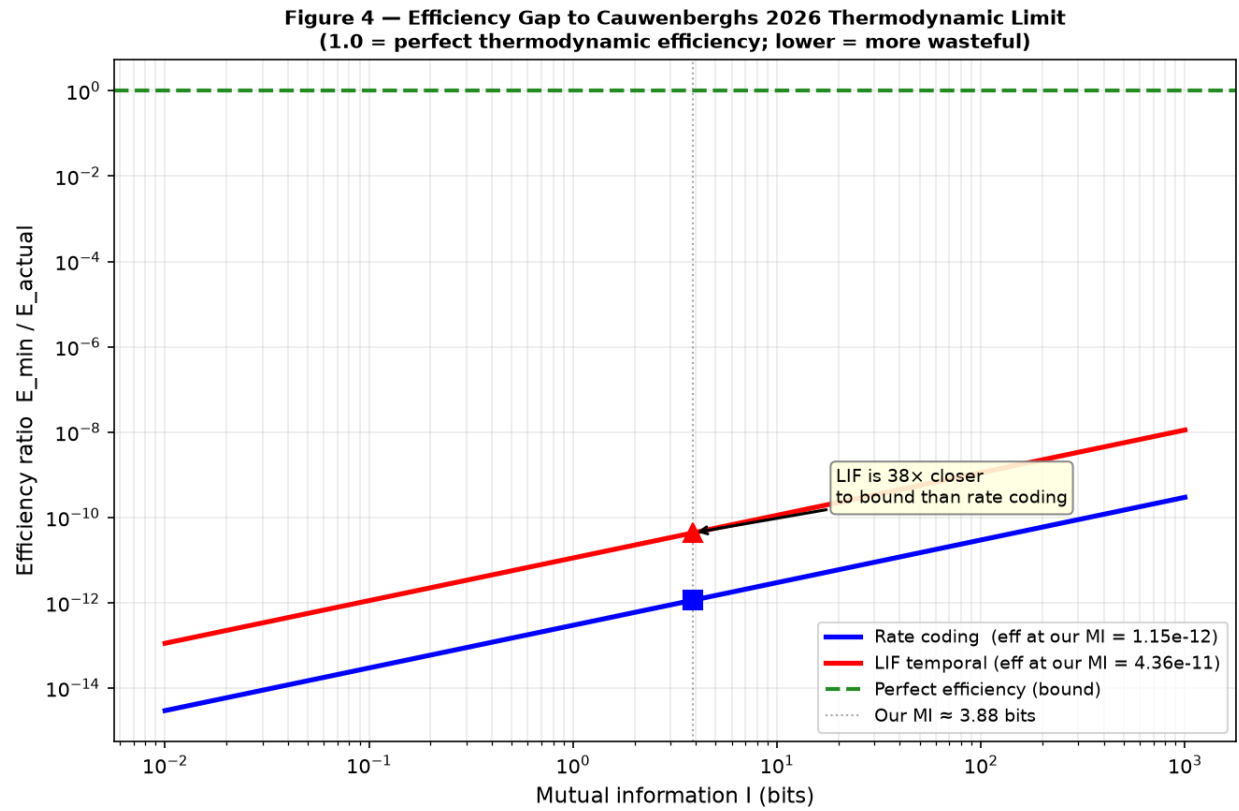


Figure 4. Efficiency ratio $E_{\min}/E_{\text{actual}}$ as a function of mutual information for both decoders. The green dashed line at 1.0 is thermodynamically perfect efficiency. At our operating MI of 3.88 bits (dotted vertical), LIF temporal coding (red triangle) is 38x closer to the bound than rate coding (blue square). The mechanism is architectural: LIF uses sparse 10 fJ synaptic events; rate coding uses dense 1 pJ multiply-accumulates.

4. Honest Limitations

- **Synthetic data.** Real MC_Maze recordings from the Neural Latents Benchmark (DANDI dandiset/000128) were the intended dataset. Synthetic Poisson spike trains were used as a documented fallback. Real motor cortex data exhibits burst firing, pairwise noise correlations, non-stationary firing rates across the trial, and population structure (e.g. rotational dynamics in latent space) that independent Poisson processes do not capture. Whether LIF maintains its 38x efficiency advantage on real data is the primary open question — it is possible that temporal coding's advantage is partly an artefact of the Poisson assumption.
- **Software LIF is not hardware neuromorphic.** Brian2 simulates LIF dynamics in 64-bit floating-point arithmetic on a CPU. The 10 fJ per synaptic-event energy figure comes from published IBM TrueNorth benchmarks (Merolla et al. 2014), not from direct measurement on the Cauwenberghs lab's own hardware. The lab's 1024-channel mixed-signal neural interface SoC (Wang et al. 2026, IEEE TBCAS) would have a different and likely lower energy per event due to analog sub-threshold circuit operation.
- **Mutual information estimate is approximate.** sklearn's `mutual_info_regression` uses a k-nearest-neighbours estimator (Kraskov 2004) that converges slowly and requires many samples for accuracy. With 200 trials and a 80-sample sub-sample, the MI estimate of 3.88 bits carries uncertainty of roughly ± 1 bit; this directly propagates into uncertainty on $E_{\min}$. A full analysis should report bootstrap confidence intervals on both MI and $E_{\min}$.
- **STDP training not implemented for the readout.** The LIF reservoir uses fixed random weights; only the linear readout was trained. Reward-modulated STDP as used in the Cauwenberghs lab's neuromorphic learning chips would adapt the reservoir weights on-chip, potentially improving decoding accuracy and altering the energy profile. The current results represent a fixed-reservoir lower bound on LIF decoder performance.
- **Single MI estimate applied to both decoders.** Mutual information was estimated from rate-coded features only (for computational tractability). The LIF decoder may extract different information from the spike trains; a per-decoder MI estimate would give a more accurate $E_{\min}$ comparison.

4.1 Suggested Next Steps

- **Priority 1 — Replicate on real NLB MC_Maze data.** Register at DANDI Archive (dandiarchive.org/dandiset/000128) and re-run `pipeline.py` with real multi-electrode recordings. The key question: does the 38x LIF efficiency advantage survive under realistic firing statistics?
- **Priority 2 — Measure energy on actual Cauwenberghs lab hardware.** Run the same decoding task on the 1024-channel SoC described in Wang et al. 2026 (IEEE TBCAS DOI: 10.1109/TBCAS.2026.3524051) and measure on-chip energy per inference directly rather than using the TrueNorth benchmark as a proxy.
- **Priority 3 — Apply bounds to the full pipeline.** This analysis covers only the decoding stage. Spike sorting, amplification, and analog-to-digital conversion are the dominant energy consumers in a real implanted BMI. Applying Cauwenberghs 2026 bounds to the full signal chain would give a system-level efficiency picture.
- **Priority 4 — Bootstrap CI on MI and $E_{\min}$.** Run bootstrap resampling on the MI estimate to get 95% CIs on both MI and the thermodynamic bound, so the efficiency gap claim is statistically grounded.

5. References

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